



In this chapter you will learn about...

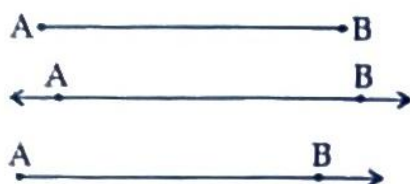
- ☐ Angle
- ☐ Pairs of angles
- ☐ Pairs of lines

INTRODUCTION

In class VI, we have learnt about lines and angles and their measurements. Let us review in brief, what we studied there and then learn about some properties of angles.

Recall that a **line segment** has two end **points**. If we extend the line segment in either direction endlessly, we get a **line**. Thus, a line has no end points. On the other hand, a **ray** has only one end point which is its starting point.

- (i) A line segment AB is denoted by $\overline{AB}$.
- (ii) A line AB is denoted by $\overleftrightarrow{AB}$.
- (iii) A ray AB is denoted by $\overrightarrow{AB}$.



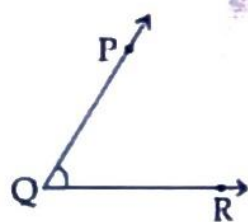
ANGLE

A figure formed by two rays with the common initial point is called an **angle**.

The common initial point is called the **vertex** of the angle and the rays forming the angle are called its **arms** or **sides**.

In the given figure, QP and QR are the two arms and Q is the vertex of the angle PQR.

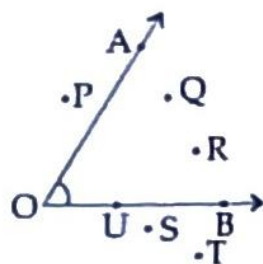
We denote it as $\angle PQR$.



Interior and Exterior of an Angle

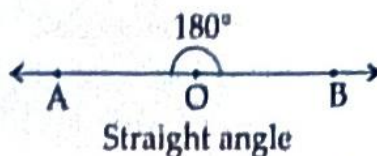
In the adjoining figure points Q and R are in the interior of $\angle AOB$.

Points A, B and U are on $\angle AOB$ and points P, S and T are in the exterior of $\angle AOB$.

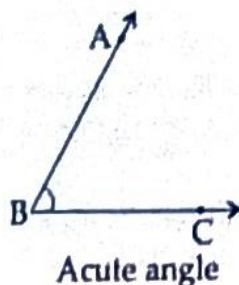


Types of Angles

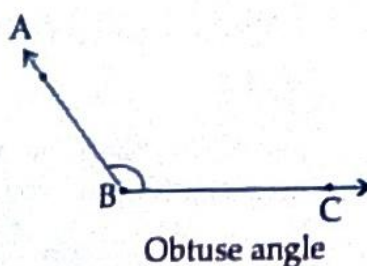
- Straight angle** : If two arms of an angle lie along the same straight line but point towards opposite directions, then the angle is called a straight angle. A straight angle always measures 180° .



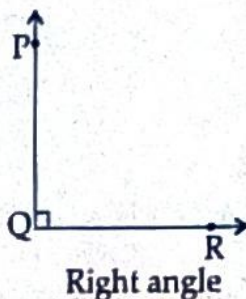
2. **Acute angle** : An angle whose measure is greater than 0° but less than 90° is called an acute angle.



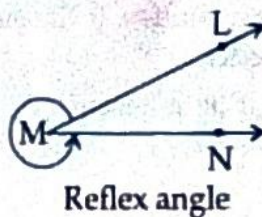
3. **Obtuse angle** : An angle whose measure is greater than 90° but less than 180° is called an obtuse angle.



4. **Right angle** : An angle whose measure is 90° is called a right angle.



5. **Reflex angle** : An angle whose measure is greater than 180° but less than 360° is called a reflex angle.



PAIRS OF ANGLES

In geometry, we often come across pairs of angles. They have been given specific names.

Adjacent Angles

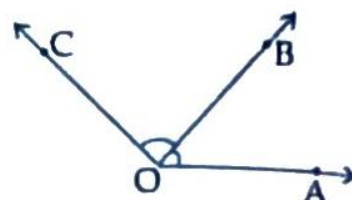
If two angles have

- (i) a common arm,
- (ii) a common vertex and
- (iii) their non-common arms lie on either side of the common arm, then the angles are called adjacent angles.

In the given figure $\angle AOB$ and $\angle BOC$ are adjacent angles.

$\overline{OB}$ is the common arm and O is the common vertex. The arms $\overline{OA}$ and $\overline{OC}$ are on the opposite sides of $\overline{OB}$.

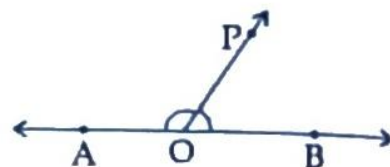
Also, $\angle AOC = \angle AOB + \angle BOC$.



Linear Pair of Angles

If a ray stands on a line, then the two adjacent angles so formed are called **linear pair of angles**. In the given figure, AB is a line and OP is a ray standing on line AB.

$\angle AOP$ and $\angle BOP$ are adjacent angles and they form linear pair of angles.



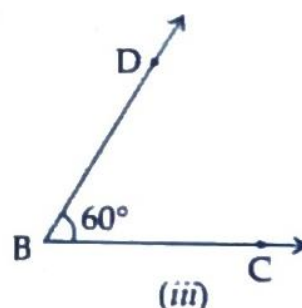
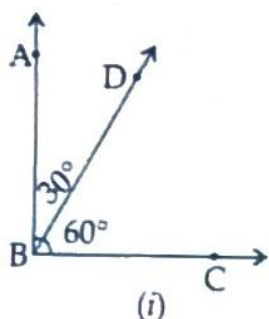
From the figure, $\angle AOP + \angle BOP = \angle AOB$

$$\Rightarrow \angle AOP + \angle BOP = 180^\circ \quad (\because \angle AOB \text{ is a straight angle})$$

It implies that, two adjacent angles will form a linear pair, if the sum of these angles is 180° .

Complementary Angles

If the sum of the measures of two angles is 90° , then the angles are called **complementary angles** and each angle is called a **complement** of the other. Thus, $\angle A$ and $\angle B$ are complementary if $\angle A + \angle B = 90^\circ$.

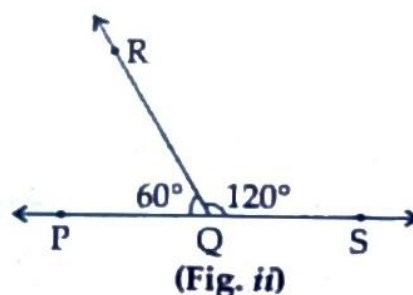
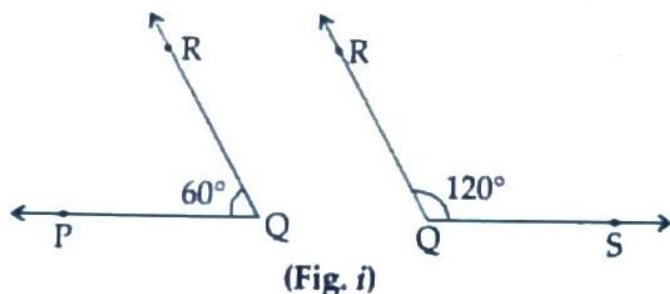


In the above figures, $\angle ABD$ and $\angle DBC$ are complementary angles, since $\angle ABD + \angle DBC = 90^\circ$.

Supplementary Angles

Two angles are said to be **supplementary angles**, if the sum of their measures is 180° and each angle is called a **supplement** of the other.

Supplementary angles may or may not be adjacent.



In the above figures (i) and (ii), $\angle PQR$ and $\angle RQS$ are supplementary angles, since $\angle PQR + \angle RQS = 180^\circ$.

Remember : The angles in a linear pair are supplementary. But supplementary angles need not form a linear pair.

Vertically Opposite Angles

When two straight lines intersect each other, then the pairs of angles so formed without any common arm are called **vertically opposite angles**.

Let PQ and RS be two lines, which intersect at O. Thus, four angles have been formed. $\angle 1$ and $\angle 3$ form a pair of vertically opposite angles, also $\angle 2$ and $\angle 4$ form another pair of vertically opposite angles.

From the figure, $\angle 1 + \angle 2 = 180^\circ$ ($\because \angle 1$ and $\angle 2$ form a linear pair)

$$\therefore \angle 1 = 180^\circ - \angle 2 \quad \dots(i)$$

Also, $\angle 2 + \angle 3 = 180^\circ$ ($\because \angle 2$ and $\angle 3$ form a linear pair)

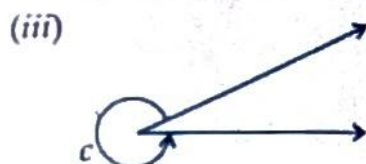
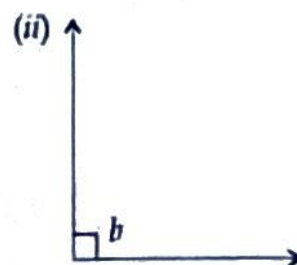
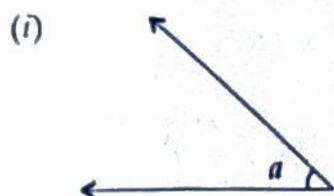
$$\angle 3 = 180^\circ - \angle 2 \quad \dots(ii)$$

From equations (i) and (ii) we get $\angle 1 = \angle 3$

Similarly, we can prove $\angle 2 = \angle 4$

Thus, if two lines intersect, then vertically opposite angles so formed are equal.

Example 1 : State which of them is acute, right, reflex and obtuse angle.



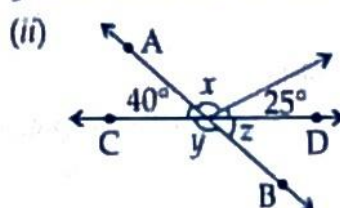
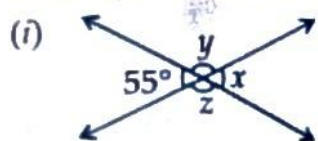
Solution : (i) $\angle a$ is an acute angle being less than 90° .

(ii) $\angle b$ is a right angle (i.e., 90°).

(iii) $\angle c$ is a reflex angle, since it is more than 180° .

(iv) $\angle d$ is an obtuse angle, since it is more than 90° but less than 180° .

Example 2 : Find the values of the angles x , y and z in each of the following :



Solution : (i) Since $\angle x$ and angle of 55° are vertically opposite angles.

$$\Rightarrow \angle x = 55^\circ$$

$\angle x$ and $\angle y$ being linear pair of angles, their sum is 180° .

$$\therefore \angle x + \angle y = 180^\circ$$

$$\Rightarrow 55^\circ + \angle y = 180^\circ$$

$$\Rightarrow \angle y = 180^\circ - 55^\circ = 125^\circ$$

Now, $\angle y = \angle z$, as they are vertically opposite angles.

$$\therefore \angle z = \angle y = 125^\circ$$

Hence, $\angle x = 55^\circ$, $\angle y = 125^\circ$ and $\angle z = 125^\circ$.

(ii) Since $\angle z$ and angle of 40° are vertically opposite angles.

$$\therefore \angle z = 40^\circ$$

Since $\angle y$ and angle of 40° form a linear pair.

$$\therefore \angle y + 40^\circ = 180^\circ$$

$$\Rightarrow \angle y = 180^\circ - 40^\circ = 140^\circ$$

AB being a straight line, therefore

$$\angle x + 25^\circ + \angle z = 180^\circ$$

$$\Rightarrow \angle x + 25^\circ + 40^\circ = 180^\circ$$

$$\Rightarrow \angle x + 65^\circ = 180^\circ$$

$$\Rightarrow \angle x = 180^\circ - 65^\circ = 115^\circ$$

Hence, $\angle x = 115^\circ$, $\angle y = 140^\circ$ and $\angle z = 40^\circ$.

(by linear pair)

Example 3 : In the adjoining figure, find $\angle COB$, if $\angle AOC = 110^\circ$.

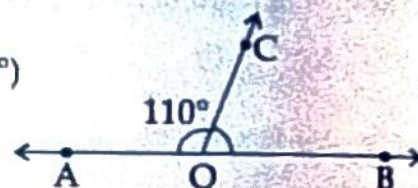
Solution : Since $\angle AOC$ and $\angle COB$ form a linear pair of angles.

$$\therefore \angle AOC + \angle COB = 180^\circ$$

$$\Rightarrow 110^\circ + \angle COB = 180^\circ \quad (\because \angle AOC = 110^\circ)$$

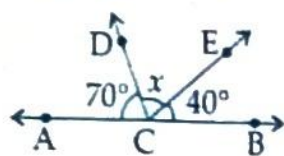
$$\Rightarrow \angle COB = 180^\circ - 110^\circ$$

$$\text{Hence, } \angle COB = 70^\circ$$



Example 4 : Find the value of x :

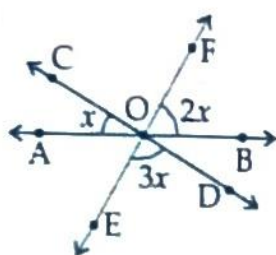
(i)



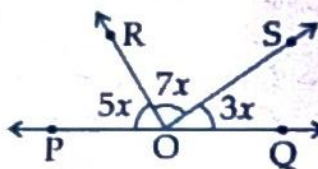
(ii)



(iii)



(iv)



Solution : (i) Since AB is a straight line and CD and CE stands on it, therefore

$$\angle ACD + \angle DCB = 180^\circ$$

(linear pair)

$$\angle ACD + (\angle DCE + \angle ECB) = 180^\circ$$

$$\Rightarrow 70^\circ + x + 40^\circ = 180^\circ$$

$$\Rightarrow 110^\circ + x = 180^\circ$$

$$\Rightarrow x = 180^\circ - 110^\circ = 70^\circ$$

(ii) Since $\angle COB$ and $\angle AOD$ are vertically opposite angles.

$$\therefore \angle COB = \angle AOD$$

$$\Rightarrow 45^\circ = x$$

$$\text{Hence, } x = 45^\circ$$

(iii) Lines CD and EF intersect at point O.

$$\therefore \angle COF = \angle EOD$$

$$\Rightarrow \angle COF = 3x$$

(Vertically opposite angles)

Also, AB is a straight line, OC and OF stand on it, therefore

$$\angle AOC + \angle COB = 180^\circ$$

(linear pair)

$$\angle AOC + (\angle COF + \angle FOB) = 180^\circ$$

$$(\because \angle COB = \angle COF + \angle FOB)$$

$$\Rightarrow x + 3x + 2x = 180^\circ$$

$$\Rightarrow 6x = 180^\circ$$

$$\text{Hence, } x = 30^\circ$$

(iv) PQ is a straight line, OR and OS stand on it, therefore

$$\angle POR + \angle ROQ = 180^\circ$$

(linear pair)

$$\therefore \angle POR + \angle ROS + \angle SOQ = 180^\circ$$

$$(\angle ROQ = \angle ROS + \angle SOQ)$$

$$\Rightarrow 5x + 7x + 3x = 180^\circ$$

$$\Rightarrow 15x = 180^\circ$$

$$\Rightarrow x = \frac{180^\circ}{15} = 12^\circ$$

$$\text{Hence, } x = 12^\circ.$$

Example 5 : An angle is equal to four times its complement. Determine its measure.

Solution : Let the measure of the given angle be x .

$$\text{Complement of } x = (90^\circ - x)$$

It is given that, required angle = $4 \times$ Complement of the angle

$$x = 4 \times (90^\circ - x)$$

$$\Rightarrow x = 360^\circ - 4x$$

$$\Rightarrow x + 4x = 360^\circ$$

$$\Rightarrow 5x = 360^\circ$$

$$\Rightarrow x = 72^\circ$$

Hence, the measure of the given angle is 72° .

Example 6 : Two supplementary angles differ by 48° . Find the angles.

Solution : Let one angle be x , then the other angle is $x + 48^\circ$.

Since x and $(x + 48^\circ)$ are supplementary angles.

$$\therefore x + (x + 48^\circ) = 180^\circ$$

$$\Rightarrow 2x + 48^\circ = 180^\circ$$

$$\Rightarrow 2x = 180^\circ - 48^\circ$$

$$\Rightarrow 2x = 132^\circ$$

$$\text{therefore, } x = 66^\circ$$

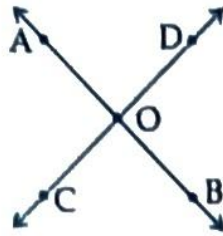
$$x = 66^\circ \text{ and } x + 48^\circ = 66^\circ + 48^\circ = 114^\circ$$

Hence, the measures of two angles are 66° and 114° .



If it were two hours later, it would be half as long until midnight as it would be if it were an hour later. What time is it now?

Example 7 : Name the four pairs of adjacent angles from the given figure :



Solution : (i) CD is a straight line and OA stands on it.

$\angle AOC$ and $\angle AOD$ are adjacent angles as they have a common vertex O, a common arm OA and the other two arms OC and OD lie on either side of the common arm OA.

(ii) AB is a straight line and OC stands on it.

$\angle AOC$ and $\angle BOC$ are adjacent angles as they have a common vertex O, a common arm OC and the other two arms OA and OB lie on either side of the common arm OC.

(iii) CD is a straight line and OB stands on it.

$\angle COB$ and $\angle DOB$ are adjacent angles as they have a common vertex O, a common arm OB and the other two arms OC and OD lie on either side of the common arm OB.

(iv) AB is a straight line and OD stands on it.

$\angle AOD$ and $\angle BOD$ are adjacent angles as they have a common vertex O, a common arm OD and the other two arms OA and OB lie on either side of the common arm OD.

Hence, four pairs of adjacent angles are,

$(\angle AOC, \angle AOD)$, $(\angle AOC, \angle BOC)$, $(\angle COB, \angle DOB)$ and $(\angle AOD, \angle BOD)$.

Example 8 : In the adjoining figure, lines AB and CD intersect at O. If $\angle 3 = 70^\circ$, find all other angles.

Solution : Since CD is a straight line and OB stands on it, therefore, $\angle 2$ and $\angle 3$ form linear pair of angles.

$$\therefore \angle 2 + \angle 3 = 180^\circ$$

$$\Rightarrow \angle 2 + 70^\circ = 180^\circ$$

$$(\because \angle 3 = 70^\circ)$$

$$\Rightarrow \angle 2 = 180^\circ - 70^\circ$$

$$\Rightarrow \angle 2 = 110^\circ$$

Since $\angle 3$ and $\angle 1$ are vertically opposite angles.

$$\therefore \angle 1 = \angle 3,$$

$$\text{But } \angle 3 = 70^\circ$$

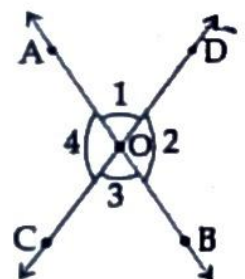
$$\Rightarrow \angle 1 = 70^\circ$$

Also, $\angle 2$ and $\angle 4$ are vertically opposite angles

$$\therefore \angle 4 = \angle 2$$

$$\Rightarrow \angle 4 = 110^\circ$$

Hence, $\angle 1 = 70^\circ$, $\angle 2 = 110^\circ$, and $\angle 4 = 110^\circ$.

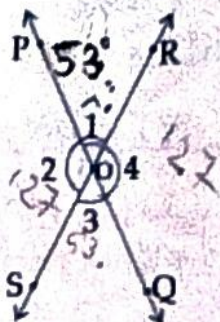


$$[\because \angle 2 = 110^\circ]$$



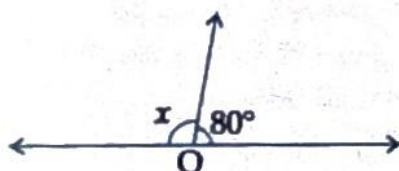
EXERCISE 11.1

- Find the complementary and supplementary angle of each of the following :
 (i) 26° (ii) 62° (iii) 9° (iv) 51° (v) 37°
- Find the complement of each of the following :
 (i) 72° (ii) 19° (iii) 88° (iv) 25°
- In the given figure, lines PQ and RS intersect at point O. If $\angle 1 = 53^\circ$, find all other angles.

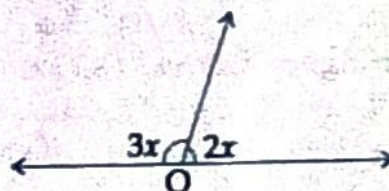


- Find the value of x in each of the following figures given below :

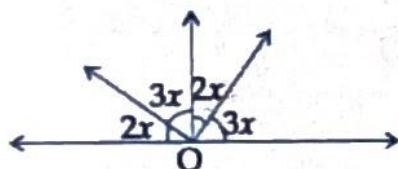
(i)



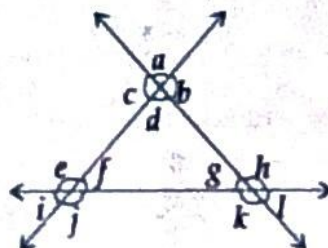
(ii)



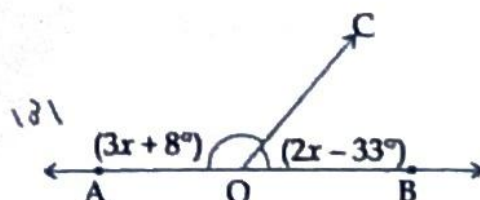
(iii)



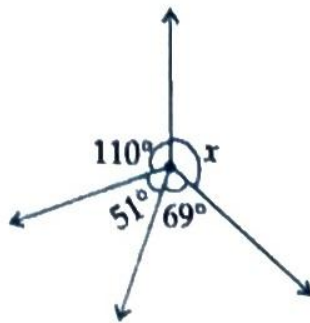
- In the given figure, name each linear pair of angles and pair of vertically opposite angles.



- In the given figure, what value of x will make AOB a straight line ?

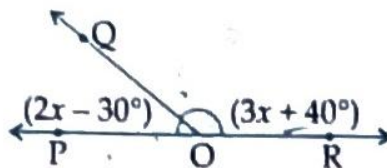


7. From the given figure, find the value of x .

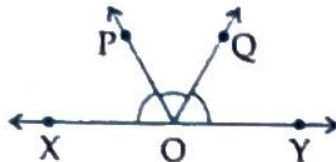


180
62

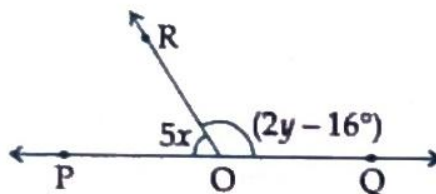
8. In the given figure, POR is a straight line and the ray OQ stands on it. Find the value of x . Also, find $\angle POQ$ and $\angle QOR$.



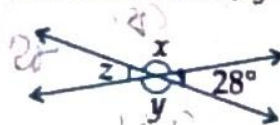
9. In the given figure, XOY is a straight line. OP and OQ stand on line XY. Write all the pairs of adjacent angles and all the linear pairs of angles.



10. In the given figure, OP and OQ are opposite rays and OR stands on PQ.



- (i) If $y = 73^\circ$, find the value of x .
 (ii) If $x = 14^\circ$, find the value of y .
11. From the adjacent figure, find the values of x , y and z .



12. In $\triangle PQR$, sides PR and QR are extended to B and A respectively. If $\angle ARB = 62^\circ$, find $\angle PRQ$ and $\angle BRQ$.

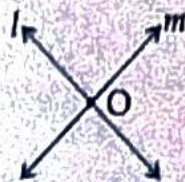


PAIRS OF LINES

We shall now study about intersecting lines, transversal and parallel lines, also angles made by a transversal with two lines.

Intersecting Lines

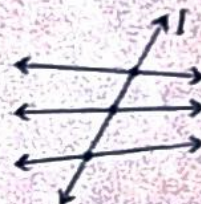
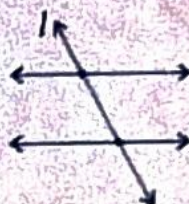
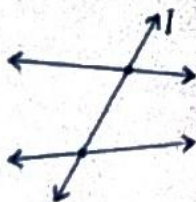
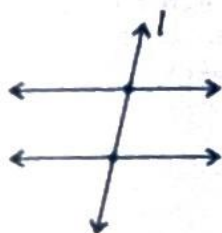
Two lines l and m intersect if they have a point in common. This common point is called the **point of intersection**. In the given figure, two lines l and m intersect at O , which is the point of intersection.



Transversal

A line which intersects two or more given lines in a plane at **different points** is called a **transversal** to the given lines.

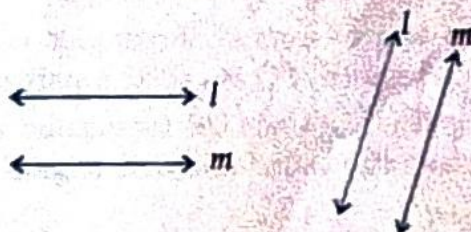
In each of the figures given below, l is a transversal.



Parallel Lines

Two lines in a plane are parallel, if they do not meet when produced infinitely in either direction.

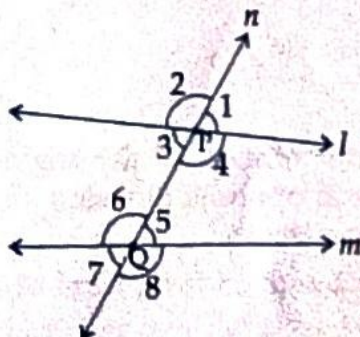
The opposite edges of a room, the opposite edges of a black board, railway lines, the opposite edges of a ruler etc. are all examples of parallel lines. In the given figures, $l \parallel m$ or $m \parallel l$.



The perpendicular distance between two parallel lines always remains the same.

Angles Formed by a Transversal with Two Lines

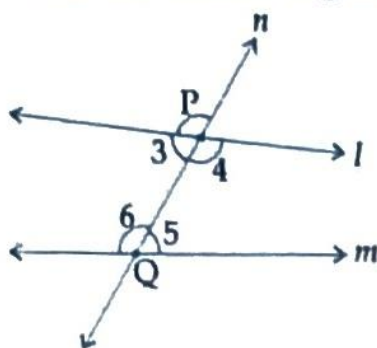
Let l and m be two lines and n be a transversal intersecting l and m at P and Q respectively. Clearly, lines l , m and n make eight angles. In the given figure, these eight angles marked 1 to 8 have their special names.



Interior angles

The angles whose arms include line segment PQ are called **interior angles**.

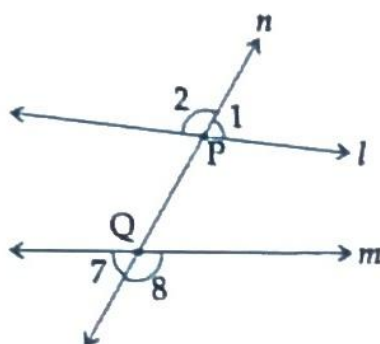
In the given figure, $\angle 3$, $\angle 4$, $\angle 5$ and $\angle 6$ are interior angles.



Exterior angles

The angles whose arms do not include the line segment PQ are called **exterior angles**.

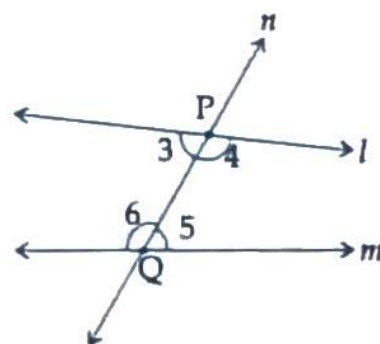
In the given figure, $\angle 1$, $\angle 2$, $\angle 7$ and $\angle 8$ are exterior angles.



Alternate interior angles

A pair of angles in which one arm of each of the angles is on the opposite sides of the transversal and the other arm is segment PQ is called a **pair of alternate interior angles**.

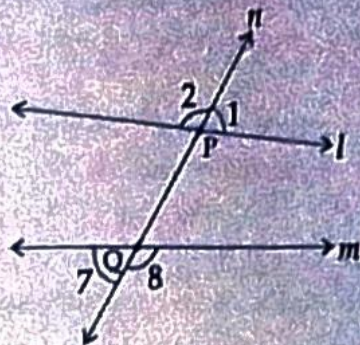
Thus, in the given figure, $\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$ form pairs of alternate interior angles. In short, alternate interior angles are also called alternate angles.



Alternate exterior angles

A pair of angles in which one arm of each of the angles is on the opposite sides of the transversal and the other arms are in opposite direction and do not include line segment PQ is called a **pair of alternate exterior angles**.

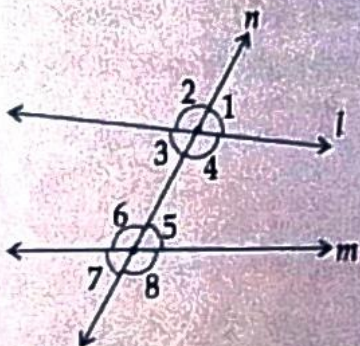
In the given figure, $\angle 1$ and $\angle 7$, $\angle 2$ and $\angle 8$ form pairs of alternate exterior angles.



Corresponding angles

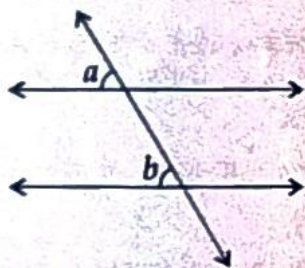
A pair of angles in which one arm of each angle of the pair is on the same side of the transversal and the other arms of the angles of the pair are directed in the same direction are known as **corresponding angles**.

In the given figure, four pairs of corresponding angles are $\angle 1$ and $\angle 5$; $\angle 2$ and $\angle 6$; $\angle 3$ and $\angle 7$; $\angle 4$ and $\angle 8$ are corresponding angles.

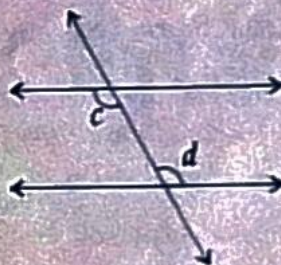


Example 1 : Name the pairs of angles in each of the following figures :

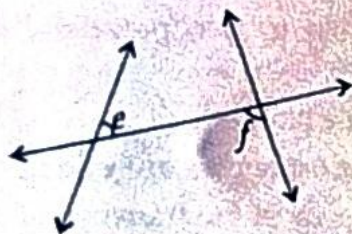
(i)



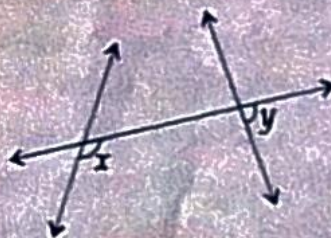
(ii)



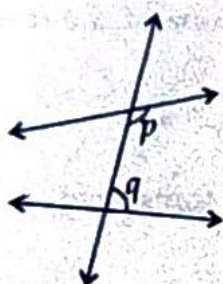
(iii)



(iv)



(v)



- Solution :**
- (i) $\angle a$ and $\angle b$ are corresponding angles because, one arm of both the angles is on the same side of the transversal and their other arms are directed in the same side.
 - (ii) $\angle c$ and $\angle d$ are alternate interior angles because the arms of each of these angles is on the opposite sides of the transversal and the other arm is common.
 - (iii) $\angle c$ and $\angle f$ are alternate interior angles because the arm of each of the angles is on the opposite sides of the transversal and the other arm is common.
 - (iv) $\angle x$ and $\angle y$ are corresponding angles because one arm of both the angles is on the same side of the transversal and other arms are directed in the same side.
 - (v) $\angle p$ and $\angle q$ are interior angles on the same side of the transversal.

Angles Made by a Transversal to Two Parallel Lines

When two lines are cut by a transversal, eight angles (four pairs) are formed. These pairs of angles have special properties, if the lines are parallel.

Let l and m be two parallel lines and n be the transversal intersecting l and m .

The transversal makes the following eight angles as shown in the figure.

$\angle 1, \angle 2, \angle 3, \angle 4, \angle 5, \angle 6, \angle 7$ and $\angle 8$.

If we measure those angles with the help of a protractor, we find that :

- (i) $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$

These are alternate interior angles.

- (ii) $\angle 1 = \angle 7$ and $\angle 2 = \angle 8$

These are alternate exterior angles.

- (iii) $\angle 1 = \angle 5; \angle 2 = \angle 6; \angle 3 = \angle 7; \text{ and } \angle 4 = \angle 8$

These are corresponding angles.

- (iv) $\angle 3 + \angle 6 = 180^\circ$ and $\angle 4 + \angle 5 = 180^\circ$

$\angle 3, \angle 6$ and $\angle 4, \angle 5$ are two pairs of interior angles.

This implies that sum of the interior angles on the same side of the transversal is 180° .

- (v) $\angle 1 + \angle 8 = 180^\circ$ and $\angle 2 + \angle 7 = 180^\circ$

$\angle 1, \angle 8$ and $\angle 2, \angle 7$ are two pairs of exterior angles on the same side of the transversal.

Thus, we observe that when a transversal is drawn to two parallel lines, the following properties of the angles are formed :

Property 1 : Each pair of corresponding angles are equal.

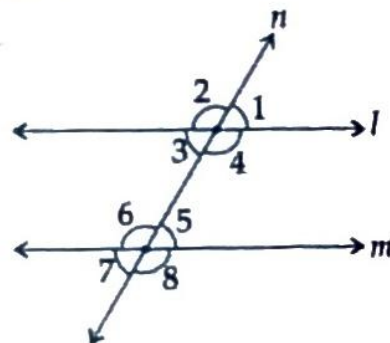
Property 2 : Each pair of alternate (interior or exterior) angles are equal.

Property 3 : Each pair of interior angles on the same side of the transversal are supplementary.

Conversely, if a transversal intersects two lines such that any one of the following conditions hold :

- (i) A pair of alternate angles are equal.
- (ii) A pair of corresponding angles are equal.
- (iii) The sum of the interior angles on the same side of the transversal is 180° .

Then, the two lines are parallel.



The following examples illustrate the application of above properties :

Example 1 : In the given figure, $DE \parallel BC$. Find the values of x and y .

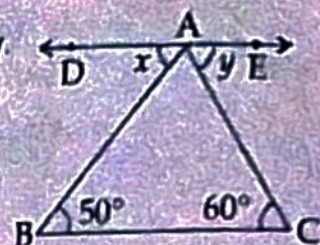
Solution : Since $DE \parallel BC$ and AB is transversal.

$$\therefore \angle x = 50^\circ \quad (\text{alternate angles})$$

Again, $DE \parallel BC$ and AC is transversal.

$$\therefore \angle y = 60^\circ \quad (\text{alternate angles})$$

Hence, $\angle x = 50^\circ$ and $\angle y = 60^\circ$.



Example 2 : In the given figure, side BC of $\triangle ABC$ has been produced to D . Line $CE \parallel BA$. If $\angle ABC = 70^\circ$ and $\angle BAC = 40^\circ$, find :

- (i) $\angle ACE$ (ii) $\angle ECD$ (iii) $\angle ACD$

Solution : (i) $CE \parallel BA$ and AC is a transversal.

$$\therefore \angle BAC = \angle ACE \quad (\text{Alternate angles})$$

$$40^\circ = \angle ACE$$

Hence, $\angle ACE = 40^\circ$

(ii) $AB \parallel EC$ and BD is a transversal intersecting the parallel lines at B and C .

$$\therefore \angle ABC = \angle ECD \quad (\text{Corresponding angles})$$

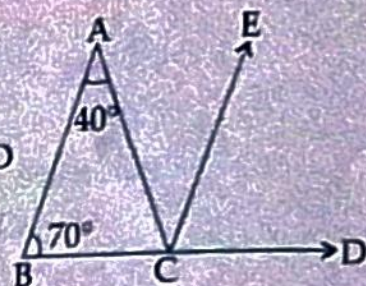
$$\Rightarrow 70^\circ = \angle ECD$$

Hence, $\angle ECD = 70^\circ$

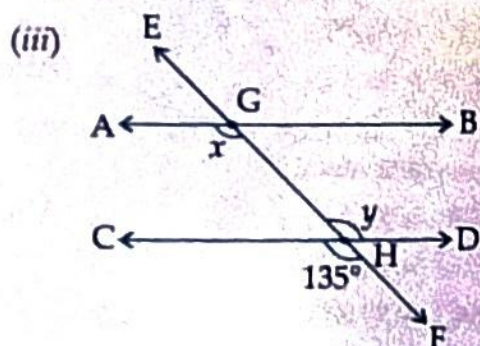
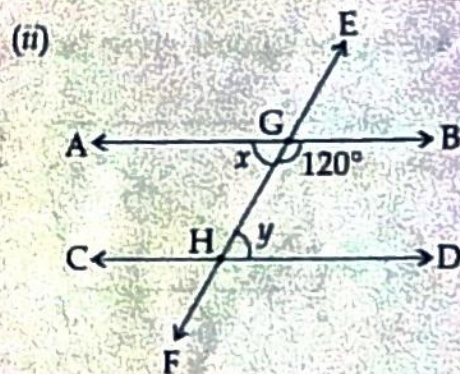
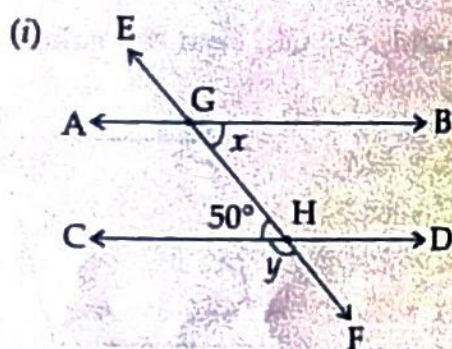
(iii) $\angle ACD = \angle ACE + \angle ECD$

$$\Rightarrow \angle ACD = 40^\circ + 70^\circ = 110^\circ$$

Hence, $\angle ACD = 110^\circ$



Example 3 : In each of the following figures, $AB \parallel CD$ and EF is a transversal intersecting the parallel lines AB and CD at G and H respectively. Find the values of x and y .



Solution : (i) $AB \parallel CD$ and EF is a transversal.

$$\therefore \angle BGH = \angle GHC$$

(Alternate angles)

Hence, $\angle x = 50^\circ$

Also, $\angle GHC + \angle CHF = 180^\circ$

(linear pair of angles)

$$\Rightarrow 50^\circ + \angle y = 180^\circ$$

$$(\because \angle GHC = 50^\circ)$$

$$\Rightarrow \angle y = 180^\circ - 50^\circ = 130^\circ$$

Hence, $\angle y = 130^\circ$

(ii) $\angle AGH$ and $\angle BGH$ form a linear pair of angles.

$$\angle AGH + \angle BGH = 180^\circ$$

$$\Rightarrow \angle x + 120^\circ = 180^\circ$$

$$[\angle BGH = 120^\circ \text{ (given)}]$$

$$\Rightarrow \angle x = 180^\circ - 120^\circ = 60^\circ$$

Hence, $\angle x = 60^\circ$

$AB \parallel CD$ and EF is a transversal.

$$\therefore \angle AGH = \angle GHD$$

(Alternate interior angles)

$$\Rightarrow \angle x = \angle y$$

$$\Rightarrow 60^\circ = \angle y$$

$$(\because \angle x = 60^\circ)$$

Hence, $\angle y = 60^\circ$

(iii) $AB \parallel CD$ and EF is a transversal.

$$\therefore \angle AGH = \angle CHF$$

(Corresponding angles)

$$\Rightarrow \angle x = 135^\circ$$

$$(\because \angle CHF = 135^\circ)$$

Again, $\angle AGH$ and $\angle GHD$ are alternate angles.

$$\therefore \angle AGH = \angle GHD$$

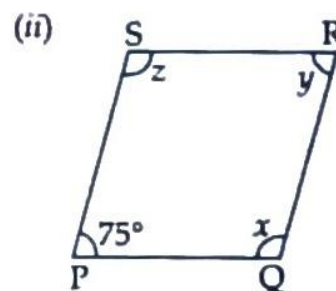
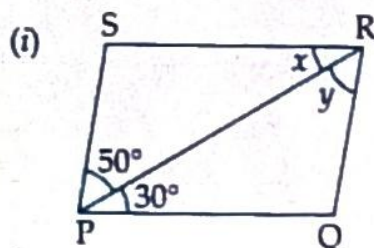
$$\Rightarrow \angle x = \angle y$$

$$\Rightarrow 135^\circ = \angle y$$

$$(\because \angle A = 135^\circ)$$

Hence, $\angle x = 135^\circ$ and $\angle y = 135^\circ$

Example 4 : In the figures given below, $PQ \parallel SR$ and $PS \parallel QR$. Find the values of unknown.



Solution : (i) $PQ \parallel SR$ and RP is a transversal.

$$\therefore \angle SRP = \angle RPQ$$

(Alternate angles)

$$\Rightarrow \angle x = 30^\circ$$

Similarly, $SP \parallel RQ$ and PR is a transversal.

$$\therefore \angle SPR = \angle PRQ$$

(Alternate angles)

$$\Rightarrow 50^\circ = \angle y$$

Hence, $\angle x = 30^\circ$ and $\angle y = 50^\circ$.

(ii) $PQ \parallel SR$ and SP is a transversal.

Sum of interior angles on the same side of transversal is 180° .

$$\therefore \angle RSP + \angle SPQ = 180^\circ$$

$$\therefore \angle z + 75^\circ = 180^\circ$$

$$\Rightarrow \angle z = 180^\circ - 75^\circ$$

$$\Rightarrow \angle z = 105^\circ$$

Now, $SP \parallel RQ$ and SR is a transversal.

Sum of interior angles on the same side of transversal is 180° .

$$\therefore \angle PSR + \angle SRQ = 180^\circ$$

$$\Rightarrow \angle z + \angle y = 180^\circ$$

$$\Rightarrow 105^\circ + \angle y = 180^\circ$$

$$\therefore \angle y = 180^\circ - 105^\circ = 75^\circ$$

Similarly, $SR \parallel PQ$ and RQ is a transversal.

Sum of interior angles on the same side of transversal is 180° .

$$\therefore \angle SRQ + \angle RQP = 180^\circ$$

$$\Rightarrow \angle y + \angle x = 180^\circ$$

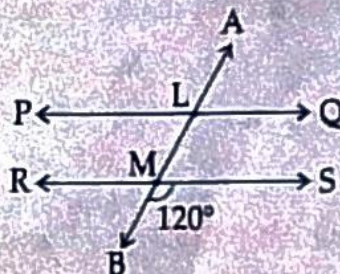
$$\Rightarrow 75^\circ + \angle x = 180^\circ$$

$$\therefore \angle x = 180^\circ - 75^\circ = 105^\circ$$

Hence, $\angle x = 105^\circ$, $\angle y = 75^\circ$ and $\angle z = 105^\circ$.

Example 5 : In the given figure, $PQ \parallel RS$ and AB is a transversal intersecting PQ and RS at L and M respectively. If $\angle SMB = 120^\circ$, find the values of remaining seven angles.

Solution : $PQ \parallel RS$ and AB is a transversal.



$$(i) \quad \angle QLM = \angle SMB \quad \text{(Corresponding angles)}$$

$$\Rightarrow \angle QLM = 120^\circ$$

$$(ii) \quad \angle BMS + \angle LMS = 180^\circ \quad \text{(Linear pair of angles)}$$

$$\Rightarrow 120^\circ + \angle LMS = 180^\circ$$

$$\Rightarrow \angle LMS = 180^\circ - 120^\circ = 60^\circ$$

$$(iii) \quad \angle PLM = \angle LMS \quad \text{(Alternate angles)}$$

$$\Rightarrow \angle PLM = 60^\circ$$

$$(iv) \quad \angle PLM + \angle LMR = 180^\circ \quad \text{(Interior angles on the same side of the transversal)}$$

$$\Rightarrow 60^\circ + \angle LMR = 180^\circ$$

$$\Rightarrow \angle LMR = 180^\circ - 60^\circ = 120^\circ$$

$$\text{Hence, } \angle LMR = 120^\circ$$

(v) $\angle ALQ$ and $\angle QLM$ being linear pair of angles,

$$\therefore \angle ALQ + \angle QLM = 180^\circ$$

$$\Rightarrow \angle ALQ + 120^\circ = 180^\circ$$

[by (i)]

$$\Rightarrow \angle ALQ = 180^\circ - 120^\circ = 60^\circ$$

(vi) $\angle ALP = \angle LMR$

(Corresponding angles)

$$\Rightarrow \angle ALP = 120^\circ$$

[by (iv)]

(vii) $\angle RMB = \angle PLM$

(Corresponding angles)

$$\Rightarrow \angle RMB = 60^\circ$$

[by (iii)]

Hence, $\angle QLM = 120^\circ$, $\angle LMS = 60^\circ$, $\angle PLM = 60^\circ$, $\angle LMR = 120^\circ$, $\angle ALQ = 60^\circ$, $\angle ALP = 120^\circ$, $\angle RMB = 60^\circ$.

Example 6 : In the given figure show that $AB \parallel EF$.

Solution : $\angle ABC = 70^\circ$ (Given)(i)

$$\angle BCD = \angle BCE + \angle ECD$$

$$\angle BCD = 30^\circ + 40^\circ$$

$$\Rightarrow \angle BCD = 70^\circ \quad \text{.....(ii)}$$

From equations (i) and (ii), we get

$$\angle ABC = \angle BCD$$

Since alternate angles are equal, therefore, $AB \parallel CD$.

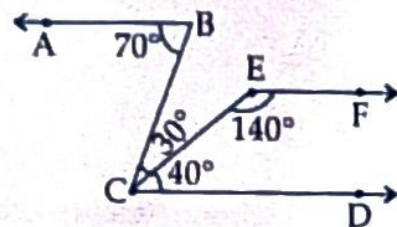
Now, $\angle FEC + \angle ECD = 140^\circ + 40^\circ = 180^\circ$.

Since, sum of the interior angles made by the transversal on its same side is 180° .

$$\therefore EF \parallel CD$$

Since, $AB \parallel CD$ and $EF \parallel CD$

$$\therefore AB \parallel EF$$



Example 7 : In the given figure line $AB \parallel CD$ and EF and GH are two transversals, find the values of a , b , c , d .

Solution : $AB \parallel CD$ and EF is a transversal.

$$\therefore \angle c = 75^\circ \quad \text{(Alternate angles)}$$

$$\text{Again } \angle c = \angle a \quad \text{(Corresponding angles)}$$

$$\therefore \angle a = 75^\circ \quad (\because \angle c = 75^\circ)$$

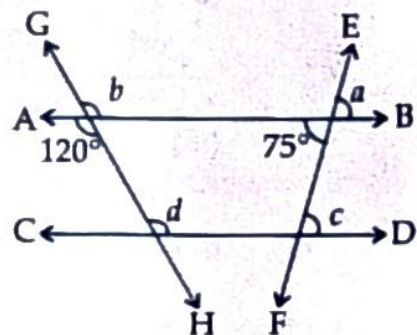
Now, $AB \parallel CD$ and GH is a transversal.

$$\therefore \angle d = 120^\circ \quad \text{(Alternate angles)}$$

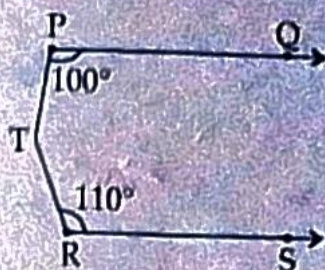
$$\text{Again } \angle b = \angle d \quad \text{(Corresponding angles)}$$

$$\therefore \angle b = 120^\circ \quad (\because \angle d = 120^\circ)$$

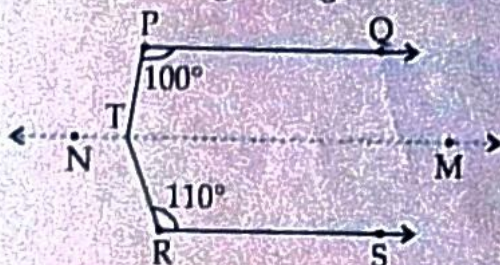
Hence, $\angle a = 75^\circ$, $\angle b = 120^\circ$, $\angle c = 75^\circ$ and $\angle d = 120^\circ$.



Example 8 : In the given figure, $PQ \parallel RS$, $\angle QPT = 100^\circ$ and $\angle TRS = 110^\circ$, then find $\angle PTR$.



Solution : Let us draw $NM \parallel PQ$ passing through T.



Since $NM \parallel PQ$ and PT is a transversal, we have

$$\angle QPT + \angle PTM = 180^\circ \quad \text{(Interior angles)}$$

$$\Rightarrow 100^\circ + \angle PTM = 180^\circ$$

$$\Rightarrow \angle PTM = 180^\circ - 100^\circ$$

$$\Rightarrow \angle PTM = 80^\circ \quad \text{---(i)}$$

Again, $NM \parallel PQ$ and $PQ \parallel RS$

$\therefore NM \parallel RS$

Since $NM \parallel RS$ and TR is a transversal, we have

$$\angle MTR + \angle TRS = 180^\circ \quad \text{(Interior angles)}$$

$$\Rightarrow \angle MTR + 110^\circ = 180^\circ$$

$$\Rightarrow \angle MTR = 180^\circ - 110^\circ$$

$$\Rightarrow \angle MTR = 70^\circ \quad \text{---(ii)}$$

Adding equations (i) and (ii), we have

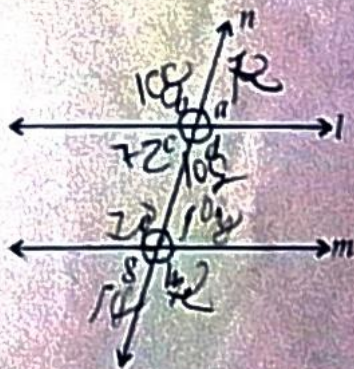
$$\angle PTM + \angle MTR = 80^\circ + 70^\circ$$

$$\therefore \angle PTR = 150^\circ$$

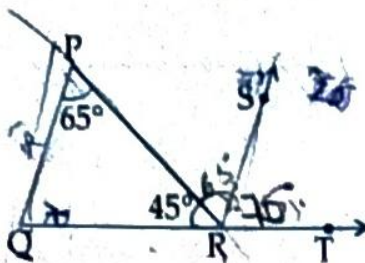


EXERCISE 11.2

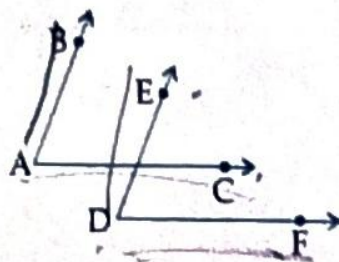
1. In the given figure, $l \parallel m$ and n is a transversal. If $\angle c = 72^\circ$, find the measure of each of the angles a, b, d, e, f, g and h .



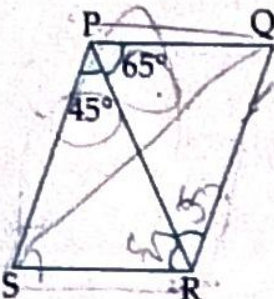
2. In the given figure, $QP \parallel RS$, $\angle P = 65^\circ$, $\angle R = 45^\circ$, then find $\angle SRT$.



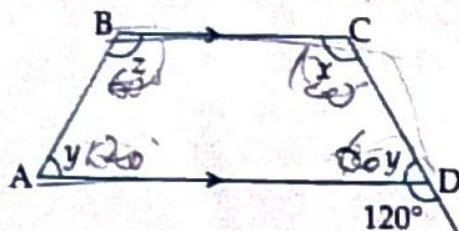
3. In the figure given below, $AB \parallel DE$ and $AC \parallel DF$, prove that $\angle BAC = \angle EDF$.



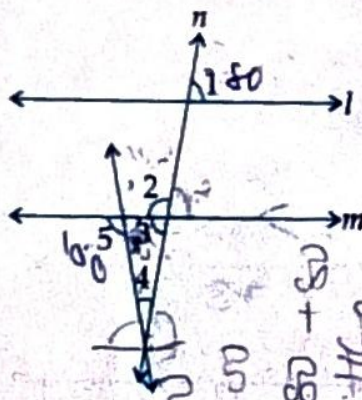
4. In the given figure, $PQ \parallel SR$ and $SP \parallel RQ$. If $\angle QPR = 65^\circ$ and $\angle SPR = 45^\circ$, find $\angle SRQ$.



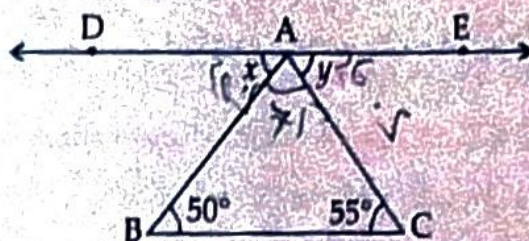
5. In the given figure, $BC \parallel AD$. Find the measures of $\angle x$, $\angle y$ and $\angle z$.



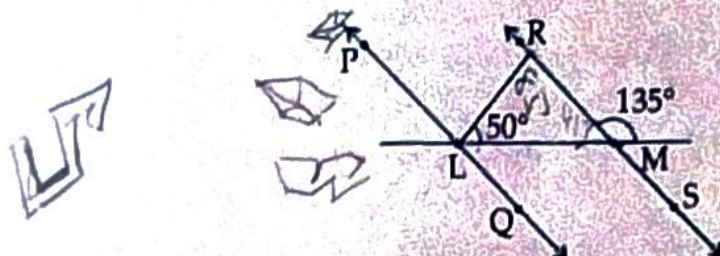
6. In the given figure, $l \parallel m$ and n is a transversal. If $\angle 1 = 80^\circ$ and $\angle 5 = 100^\circ$, find the measures of $\angle 2$, $\angle 3$ and $\angle 4$.



7. In the given figure, $BC \parallel DE$. Find the values of x and y .

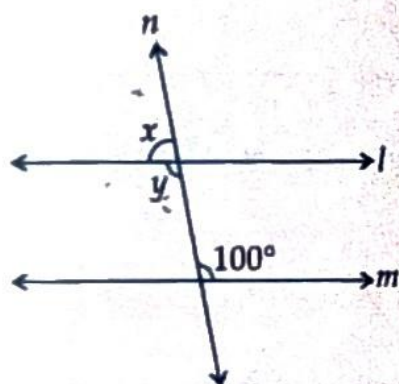


8. In the given figure, $PQ \parallel RS$, find the measure of $\angle LRM$.

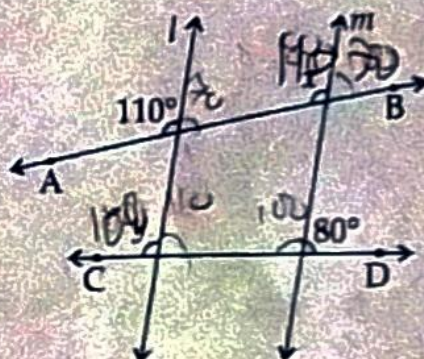


9. In each of the following figures, $l \parallel m$, find the values of x and y .

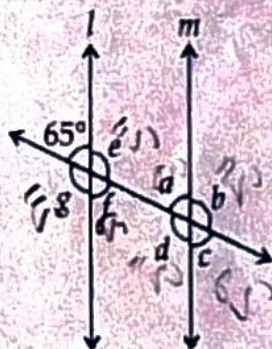
(i)



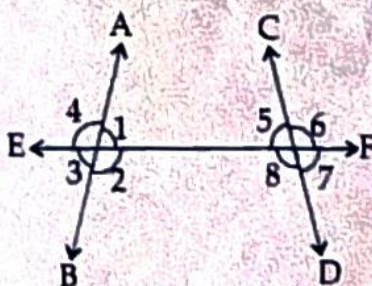
(ii)



10. In the given figure, $l \parallel m$, find the unknown angles.



11. Using the given figure, name the following angles :



(i) Corresponding angles

(ii) Alternate interior angles

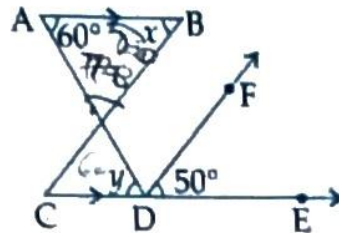
(iii) Alternate angle of $\angle 2$

(iv) Angle corresponding to $\angle 7$

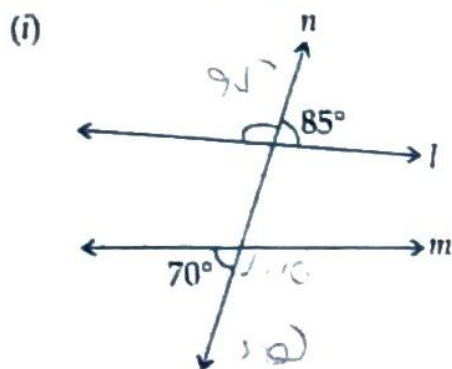
(v) Pairs of interior angles on the same side of the transversal.

12. In the given figure $AB \parallel CE$ and $DF \parallel CB$, find the values of x and y .

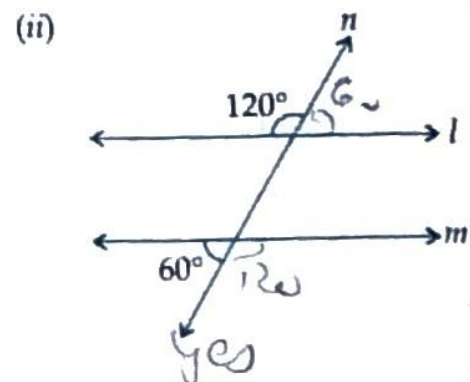
$60 = x$
 $60 = y$



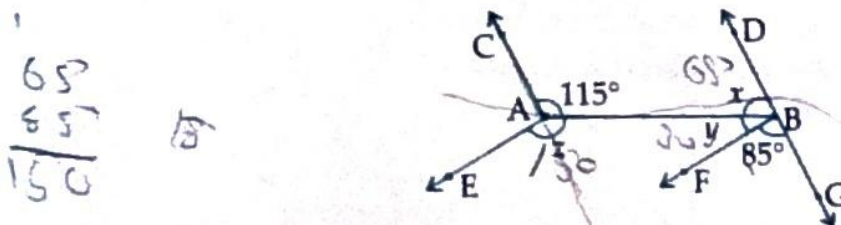
13. In the following figures, a transversal n cuts two lines l and m .



Is $l \parallel m$?



14. In the given figure, $AC \parallel BD$ and $AE \parallel BF$. Find the values of x , y and z .



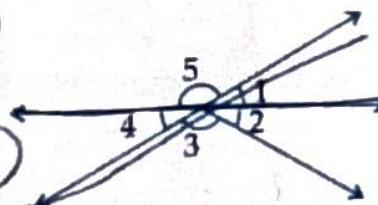
15. In the adjoining figure, indicate which pairs of angles are :

(i) linear pairs of angles

(ii) vertically opposite angles

(1) (5, 1), (4, 3), (3, 2)

(2) (5, 4), (5, 1), (1, 4)

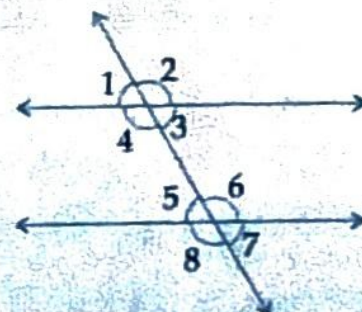




FACTS TO REMEMBER

- A line segment has two end points, a ray has only one end point and a line has no end points.
- Two angles having a common vertex, one common side and which do not overlap are called adjacent angles.
- The measure of two complementary angles add up to 90° .
- The measure of two supplementary angles add up to 180° .
- If two lines intersect, then two pairs of angles formed without a common arm are called vertically opposite angles.
- Vertically opposite angles are always equal.
- When two lines drawn on sheet of paper do not meet, however far produced, they are called parallel lines.
- A transversal is a line that intersects two or more lines in a plane at distinct points.
- A transversal gives rise to several types of angles :

Types of angles	Angles shown
Interior angles	$\angle 3, \angle 4, \angle 5, \angle 6$
Exterior angles	$\angle 1, \angle 2, \angle 7, \angle 8$
Corresponding angles	$\angle 1$ and $\angle 5$; $\angle 4$ and $\angle 8$; $\angle 2$ and $\angle 6$; $\angle 3$ and $\angle 7$.
Alternate interior angles	$\angle 3$ and $\angle 5$; $\angle 4$ and $\angle 6$
Alternate exterior angles	$\angle 1$ and $\angle 7$; $\angle 2$ and $\angle 8$
Interior angles on the same sides of transversal	$\angle 4$ and $\angle 5$; $\angle 3$ and $\angle 6$



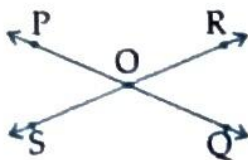
- When a transversal intersects two parallel lines :
 - (i) each pair of corresponding angles are equal.
 - (ii) each pair of alternate (interior or exterior) angles are equal.
 - (iii) each pair of interior angles on the same side of transversal are supplementary.
- Linear pair of angles are supplementary.



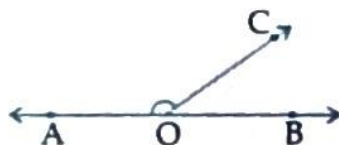
MULTIPLE CHOICE QUESTIONS

Choose the correct alternatives in each of the following :

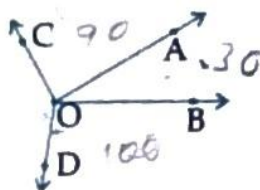
1. When two lines intersect at a point, number of pairs of adjacent angles formed is
(a) 4 (b) 3 (c) 2 (d) 6
2. The complement of an angle of 36° is
(a) 34° (b) 54° (c) 64° (d) 144°
3. The supplement of an angle of 68° is
(a) 22° (b) 142° (c) 122° (d) 112°
4. The supplement of an angle is 70° , then the complement of that angle is
(a) 32° (b) 162° (c) 52° (d) does not exist
5. The complement of an angle is 26° , then the supplement of that angle is
(a) 64° (b) 154° (c) 116° (d) does not exist
6. In the given figure, two straight lines intersect at point O, if $\angle POS = 45^\circ$, then $\angle QOR$ is equal to



- (a) 135° (b) 55° (c) 155° (d) 45°
7. AOB is a straight line, a ray OC stands on it. If $\angle AOC = 145^\circ$, then $\angle BOC$ is equal to



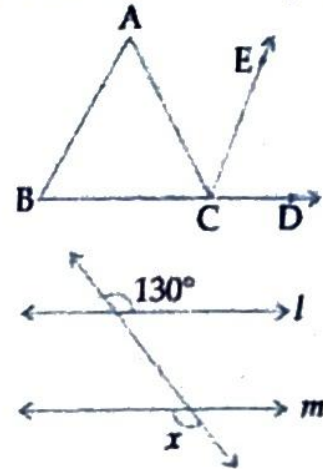
- (a) 55° (b) 45° (c) 35° (d) 145°
8. In the given figure, rays OA, OB, OC and OD are such that $\angle AOB = 30^\circ$, $\angle AOC = 90^\circ$, $\angle BOD = 100^\circ$ and $\angle COD = x$, then the value of x is



- (a) 120° (b) 140° (c) 110° (d) 135°

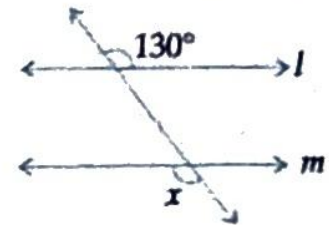
9. In the given figure, $AB \parallel EC$, $\angle A = 60^\circ$ and $\angle ECD = 70^\circ$, then $\angle ACB$ is equal to

- (a) 50°
(b) 60°
(c) 70°
(d) 40°



10. In the given figure, $l \parallel m$, the value of x is

- (a) 50°
(b) 130°
(c) 120°
(d) 100°



11. An angle is equal to 5 times its complement, then its measure is

- (a) 25° (b) 50° (c) 75° (d) 60°

12. Two angles can be supplementary, if both of them are

- (a) acute angles (b) obtuse angles (c) right angles (d) straight angles



MENTAL MATHS CORNER

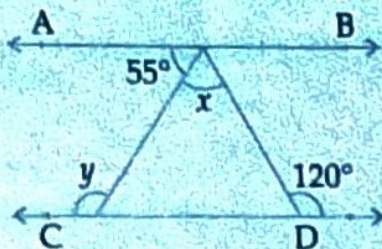
Fill in the blanks :

- If two angles of a linear pair are equal, then measure of each angle is
- If the magnitude of an angle is same as its complement, then measure of the angle is
- If the magnitude of an angle is same as its supplement, then the angle is
- Two angles are such that one of the angles is $\frac{4}{5}$ of its supplement, then the angle is and its supplement is
- Two angles forming a linear pair are
- If two adjacent angles are supplementary, they form a
- If two lines intersect at a point, then the are always equal.
- An angle is greater than 45° , then its complementary angle is than 45° .
- An angle is $\frac{2}{3}$ of its complement, then the angle is and its complement is

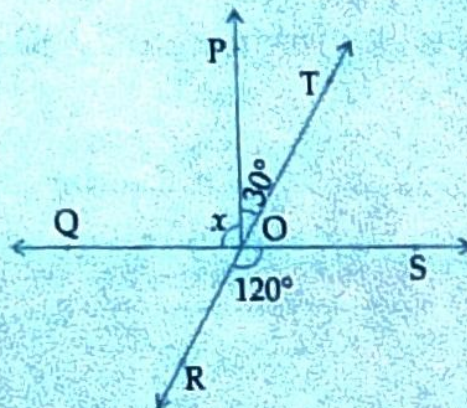


HOTS QUESTIONS

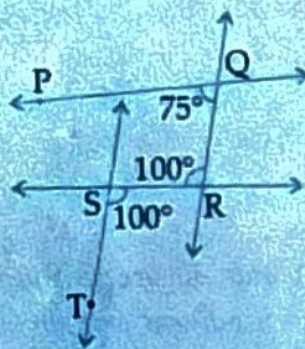
1. In the adjoining figure, if $AB \parallel CD$, find the measures of x and y .



2. Find the value of x from the given figure.



3. Which lines are parallel? Give reasons.





ANSWERS

Exercise 11.1

- (i) 64° , 154° (ii) 28° , 118° (iii) 81° , 171°
(iv) 39° , 129° (v) 53° , 143°
- (i) 18° (ii) 71° (iii) 2° (iv) 65°
- $\angle 2 = 127^\circ$, $\angle 3 = 53^\circ$, $\angle 4 = 127^\circ$
- (i) 100° (ii) 36° (iii) 18°
- Linear pair : (a, c) , (c, d) , (d, b) , (a, b) , (e, f) , (i, e) , (i, j) , (j, f) , (g, h) , (g, k) , (k, l) , (l, h) .
Vertically opposite angles : (a, d) , (c, b) , (e, j) , (i, f) , (g, l) , (h, k) .
- 41° 7. 130° 8. 34° , 38° , 142°
- Adj. angles : $(\angle XOP, \angle POQ)$, $(\angle POQ, \angle QOY)$, $(\angle XOP, \angle POY)$, $(\angle XOQ, \angle QOY)$.
Linear pair : $(\angle XOP, \angle POY)$, $(\angle XOQ, \angle QOY)$.
- (i) $x = 10^\circ$ (ii) $y = 63^\circ$ 11. 152° , 152° , 28° .
- 62° , 118°

Exercise 11.2

- 72° , 108° , 108° , 108° , 72° , 72° , 108° 2. 70°
- 110°
- 120° , 60° , 120° 6. 100° , 80° , 20°
- $x = 50^\circ$, $y = 55^\circ$ 8. 85°
- (i) $x = 80^\circ$, $y = 100^\circ$ (ii) $x = 110^\circ$, $y = 100^\circ$
- $a = 65^\circ$, $b = 115^\circ$, $c = 65^\circ$, $d = 115^\circ$, $e = 115^\circ$,
 $f = 65^\circ$, $g = 115^\circ$
- (i) $(4, 5)$, $(1, 6)$, $(3, 8)$, $(2, 7)$ (ii) $(1, 8)$, $(2, 5)$
(iii) $\angle 5$ (iv) $\angle 2$ (v) $(1, 5)$, $(2, 8)$
- $x = 50^\circ$, $y = 60^\circ$ 13. (i) No (ii) Yes
- $x = 65^\circ$, $y = 30^\circ$, $z = 150^\circ$

15. (i) $(1, 5)$, $(4, 5)$ (ii) $(4, 1)$

Multiple Choice Questions

- | | | | |
|--------|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (d) | 4. (d) |
| 5. (c) | 6. (d) | 7. (c) | 8. (b) |
| 9. (a) | 10. (b) | 11. (c) | 12. (c) |

Mental Maths Corner

- 90° 2. 45° 3. 90°
- 80° , 100° 5. Supplementary
- Linear pair
- Vertically opposite angles
- Less 9. 36° , 54° 10. 72° , 108°
- 50° 12. 0°

Review Exercise

- (i) $(1, 2)$, $(2, 3)$, $(3, 4)$, $(1, 4)$, $(6, 5)$, $(6, 7)$,
 $(7, 8)$, $(5, 8)$
(ii) $(2, 4)$, $(1, 3)$, $(6, 8)$, $(5, 7)$
- $\angle 2 = 70^\circ$, $\angle 1 = 110^\circ$, $\angle 3 = 110^\circ$
- $\angle 1 = 115^\circ$, $\angle 2 = 65^\circ$, $\angle 3 = 65^\circ$, $\angle 4 = 115^\circ$,
 $\angle 5 = 65^\circ$
- (i) No (ii) Yes (iii) Yes
- 120° 6. $\angle x = 110^\circ$, $\angle y = 70^\circ$, $\angle r = 110^\circ$
- 30°
- $\angle 1 = 72^\circ$, $\angle 2 = 108^\circ$, $\angle 3 = 72^\circ$, $\angle 4 = 108^\circ$,
 $\angle 5 = 72^\circ$, $\angle 6 = 108^\circ$, $\angle 7 = 72^\circ$, $\angle 8 = 108^\circ$

HOTS Questions

- $x = 65^\circ$, $y = 125^\circ$ 2. 90°
- QR $\parallel$ ST